Decentralized Formation Control For Multiple UAVs Based On Leader-Following Consensus With Time-Varying Delays

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***Abstract*—Combined with the decentralized method, a leader-following consensus protocol is studied with time-varying communication and self delays for multi-UAV formation control systems. By using Lyapunov and the linear matrix inequality methods, it can be proven that all the nodes can track the reference values asymptotically and the maximal allowable upper bounds of delays can be obtained. At last, simulation results of four UAVs formation are given to demonstrate the effectiveness and application value of the proposed method.**

Keywords—UAV; Formation; Leader-following consensus; Decentralized; LMI

I. INTRODUCTION

Recent years, the formation control of UAVs(Unmanned Aerial Vehicles) has attracted more and more attention[1][2][3]. UAV formation flight is that multiple UAVs flight at a given speed and some kind of formation. Furthermore, they cooperate with each other to solve some tasks which are difficult for the traditional single aircraft[4]. Paper[1] proposed a formation control method based on autopilot. Based on this, lead and wing aircrafts formation controller was designed. Justh[2] investigated the two-aircraft formation control based on Frenet-Serret formula. In [3], the model of relative motion of formation flight was established in the path coordinate frame of the wing UAVs. The distance of wing UAV and lead UAV was adjusted by PID algorithm.

Distributed consensus problem of multi-agent systems has obtained increasing attention because of its broad applications to cooperative control areas including unmanned aerial vehicles, formation control, flocking, mobile robots, etc. Different from centralized control strategy, without global communication and centralized control, decentralized control only depends on local communication so that the control purposes can be achieved. The consensus protocol is a kind of local controller which can carry out the decentralized control strategy[5][6]. Ren[7] applied the consensus protocol to second-order integral model to implement decentralized formation maintain of multi-agent system. Furthermore, in [8], Ren made use of the leader-following consensus protocol as the state estimator and controller to achieve the multi-robot formation control. In actual flight, there are some communication delays due to communication affected by the transmission speed, network congestion and other factors. So the researches of consensus protocol with time delays have important application value. In this paper, we first analyzed the model of formation control of UAVs. By employing the Lyapunov function, we put forward the leader-following consensus protocol with time-varying communication and self delays. Then, the simulation case implemented the multi-UAV formation control which verify the effectiveness of model and control strategy.

II. PROBLEM STATEMENT

A. UAV kinematic models

Consider the particle model of UAVs in the two-dimensional plane:

$$\begin{pmatrix} \dot{x}_i \\ \dot{y}_i \\ \dot{\theta}_i \end{pmatrix} = \begin{pmatrix} \cos\theta_i \\ \sin\theta_i \\ 0 \end{pmatrix} v_i + \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \omega_i \qquad i \in \{1, \cdots, n\} \tag{1}$$

where $i \in \{1, \cdots, n\}$ are the numbers of UAVs. (x_i, y_i) and θ_i denote the positions and the heading angle of UAV i. The input variables are linear velocity v_i and angular rate ω_i.

Let $r_i = [x_i, y_i]$, $u_i = [u_{xi}, u_{yi}]$, though feedback linearization of (1), we can get the first order integral form of system model.

$$\begin{bmatrix} \dot{x}_i \\ \dot{y}_i \end{bmatrix} = \begin{bmatrix} u_{xi} \\ u_{yi} \end{bmatrix} \tag{2}$$

where u_i is the virtual control input. The relationship between u_i and v_i, ω_i is

$$v_i = \sqrt{u_{xi}^2 + u_{yi}^2}$$

$$w_i = \frac{u_{xi}\dot{u}_{yi} - u_{yi}\dot{u}_{xi}}{u_{xi}^2 + u_{yi}^2}$$

B. Formation control

The multi-UAV formation control is to make the multiple UAVs fly at a certain speed and given formation. Take four UAVs system as an example, as shown in Fig.1. a square. Its center $r_c^d = (x_c^d, y_c^d)$ moves at the expected path. Assuming the path planning completed, UAVs only need fly along it. According to the unchanging of relative position, we can calculate desired coordinates of each UAV, $r_i^d = (x_i^d, y_i^d)$.

$$\begin{bmatrix} x_i^d \\ y_i^d \end{bmatrix} = \begin{bmatrix} x_c^d \\ y_c^d \end{bmatrix} + \begin{bmatrix} \tilde{x}_{ic} \\ \tilde{y}_{ic} \end{bmatrix} \tag{3}$$

where $\tilde{r}_{ic} = (\tilde{x}_{ic}, \tilde{y}_{ic})$ is the error from a desired position. If each UAV r_i can accurately track its desired coordinate r_i^d and the tracking errors converge to zero, the formation control can be achieved.

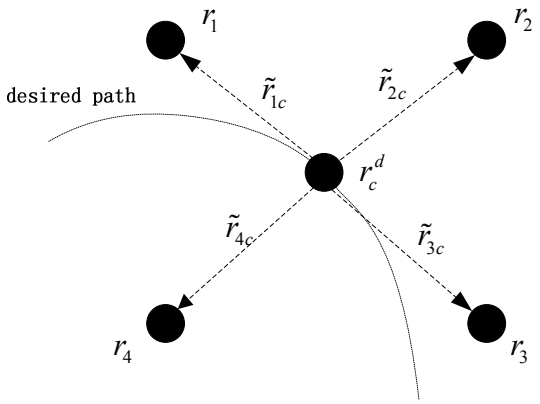


Fig.1 Structure of multiple UAVs formation

C. Graph Theory

Let a weighted directed graph $G(V, E, A)$ be the description of the interactions between the multi-agent with a set of nodes $V = \{v_1, ..., v_n\}$, a set of edges $E \subset V \times V$ and a weighted adjacency matrix $A = [a_{ij}]$ with nonnegative adjacency elements a_{ij}. The node indexes belong to a finite index set $I = \{1, ..., n\}$. An edge $e_{ij} = (v_i, v_j)$ denotes that the head of the edge v_i can obtain the information from the end of the edge v_j. The adjacency elements associated with the edges of the graph are positive, i.e., $e_{ij} = (v_i, v_j) \in E$ if and only if $a_{ij} > 0$. Moreover, we assume $a_{ii} = 0$ for all $i \in I$. The set of neighbors of node v_i is denoted by $N_i = \{v_j \in V | (v_i, v_j) \in E\}$. n the directed graph of order n, the number of edges that set out from node v_i is called out-degree which is defined as $d_i = \sum_{j \neq i} a_{ij}$. The out-degree matrix of directed graph G is defined as $D = diag(d_1, d_2, ..., d_n)$.

III. ALGORITHM DESIGN

A. Formation control strategy

The control strategies of UAV formation can be divided into two cases: centralized[10] and decentralized control strategy[11].

1) Centralized control strategy

There is one centralized control platform which can achieve the information of each UAV like position, speed, posture and moving target. In the platform, centralized computer know the information of the whole system and can choose the optimization algorithm for best control performance. However, it is inevitable to require a large amount of information exchange so that it is strict on the centralized computer. If it is out of work, the whole system will be affected.

2) decentralized control strategy

The centralized control platform is not required in decentralized control strategy. Each UAV only communicates with its neighbor UAVs, that is, all the UAVs are at the same level. When some UAVs go wrong, they will not affect the rest. This strategy has good robustness and fault-tolerant ability.

B. Leader-following consensus protocol

The information states with agent dynamics can be described as the integral form of first-order as following:

$$\dot{x}_i(t)=u_i(t),\quad i\in I \tag{4}$$

where the input $u_i(t)$ is called a protocol with topology G if it only depends on the states of node υ_i and its neighbors. Consensus refers to that without global communication and centralized control, the states of the multi-agent (such as location, speed, etc.) converges to a fixed value, only depending on local communication between the agent neighbors[5]. The leader-following consensus protocol can make the states of agents converge to a specified value which is called reference State.

It is well known that time delays which may destabilize the formation control system are inevitable[9]. It is necessary to investigate the consensus problem with time delays. In this paper, the following protocol is considered to study the formation control problem.

$$\begin{aligned}u_i(t)=&\sum_{j\in N_i}a_{ij}\left(x_j(t-\tau_{ij}(t))-x_i(t-T_i(t))\right)\\&-\alpha_i\left(x_i(t-T_i(t))-x^r\right)\end{aligned} \tag{5}$$

where $\tau_{ij}(t),T_i(t)$ are different communication and self delays. If the agent i can be access to the reference state directly, it is known as the leader and $\alpha_i>0$, otherwise $\alpha_i=0$. $\tau_{ij}(t),T_i(t),i,j\in I$ are unknown and piecewise continuous. For the sake of convenience, let $\tau_{ij}(t)\in\{d_k(t):k=1,2,...,m\}$, $T_i(t)\in\{\Gamma_r(t):r=1,2,...,n\}$ and $d_k(t),\Gamma_r(t)$ to satisfy $0\le d_k(t)\le h_{dk},0\le\Gamma_r(t)\le h_{\Gamma r}$, where $h_{dk},h_{\Gamma r}$ are constants to be determined. Rewrite (5)in the following matrix form:

$$\begin{aligned}\dot{x}(t)=&\sum_{k=1}^{m}A_kx(t-d_k(t))-\sum_{r=1}^{n}D_rx(t-\Gamma_r(k))\\&-Cx(t)+C\begin{bmatrix}x^r\\x^r\\\vdots\\x^r\end{bmatrix}\end{aligned}$$

where $C=\sum_{r=1}^{n}C_r(\alpha^{(r)})=diag(\alpha_1,\alpha_2,...,\alpha_n)$. The definition of $D_r=diag\left(d_i^{(r)}\right)$, $A_k=\left[a_{ij}^{(k)}\right]$ are as following:

$$a_{ij}^{(k)}=\begin{cases}a_{ij}, & j\ne i,d_k(\cdot)=\tau_{ij}(\cdot)\\0, & j\ne i,d_k(\cdot)\ne\tau_{ij}(\cdot),\\0, & j=i\end{cases}$$

$$d_i^{(r)}=\begin{cases}d_i & \Gamma_r=T_i\\0 & \Gamma_r\ne T_i\end{cases},\alpha_i^{(r)}=\begin{cases}\alpha_i & \Gamma_r=T_i\\0 & \Gamma_r\ne T_i\end{cases}$$

In this part, we assume the time delays satisfy:

(A1) $0\le h_{dk}<h_d,0\le h_{\Gamma r}<h_\Gamma$

So we have the following Theorem.

Theorem 1. Assume that the fixed connected topology of G satisfies (A1). If there exists $h_d>0$ and $h_\Gamma>0$ satisfying the following LMI, system (5) will track the reference state asymptotically

$$\begin{bmatrix}\Theta & PA_1 & \cdots & PA_m & -PG_1 & \cdots & -PG_n & \Phi\\ * & -Q_1 & \cdots & 0 & 0 & \cdots & 0 & \Omega_{A1}\\ \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots & \vdots\\ * & 0 & \cdots & -Q_m & 0 & \cdots & 0 & \Omega_{Am}\\ * & 0 & \cdots & 0 & -R_1 & \cdots & 0 & \Omega_{D1}\\ \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots & \vdots\\ * & 0 & \cdots & 0 & 0 & \cdots & -R_n & \Omega_{Dn}\\ * & * & \cdots & * & * & \cdots & * & \Psi\end{bmatrix}<0 \tag{6}$$

where $P>0$, $Q_1>0,\cdots,Q_m>0$, $R_1>0,\cdots,R_n>0$ are Positive definite matrixes with appropriate dimension and

$$Q=\sum\nolimits_{k=1}^{m}Q_k,R=\sum\nolimits_{r=1}^{n}R_r,$$

$$\Theta=\left(A-D-C\right)^TP+P\left(A-D-C\right),$$

$$\Phi=\left(A^T-D-C^T\right)\left(h_d^2Q+h_\Gamma^2R\right),$$

$$G_r=D_r+C_r\quad r=1,2,...,n$$

$$\Omega_{Ak}=A_k^T\left(h_d^2Q+h_\Gamma^2R\right)\quad k=1,2,...,m,$$

$$\Omega_{Dr}=-G_r^T\left(h_d^2Q+h_\Gamma^2R\right)\quad r=1,2,...,n$$

$$\Psi=-\left(h_d^2Q+h_\Gamma^2R\right)$$

Proof:

Let $y_i(t)=x_i(t)-x^r, i=1,2,3,...,n$, we have $y(t)=\left[y_1(t),y_2(t),...,y_n(t)\right]^T$ and the matrix form becomes:

$$\dot{y}(t)=\sum_{k=1}^{m}A_k y(t-d_k(t))-\sum_{r=1}^{n}D_r y(t-\Gamma_r(k))-Cy(t) \quad (7)$$

Thus, the consensus problem of original system(5)is equivalent to the stability of error system(7)at the zero solution.

Construct the Lyapunov function

$$V(y(t))=y^T(t)Py(t)+h_d\sum_{k=1}^{m}\int_{-h_d}^{0}\int_{t+\theta}^{t}\dot{y}^T(s)Q_k\dot{y}(s)dsd\theta$$
$$+h_\Gamma\sum_{r=1}^{n}\int_{-h_\Gamma}^{0}\int_{t+\theta}^{t}\dot{y}^T(s)R_k\dot{y}(s)dsd\theta$$

Its derivation along the solution of system (7) is

$$\dot{V}=\dot{y}^T(t)Py(t)+y^T(t)P\dot{y}(t)$$
$$+\sum_{k=1}^{m}h_d^2\dot{y}^T(t)Q_k\dot{y}(t)-h_d\sum_{k=1}^{m}\int_{t-h_d}^{t}\dot{y}^T(s)Q_k\dot{y}(s)ds$$
$$+\sum_{r=1}^{n}h_\Gamma^2\dot{y}^T(t)R_r\dot{y}(t)-h_\Gamma\sum_{r=1}^{n}\int_{t-h_\Gamma}^{t}\dot{y}^T(s)R_r\dot{y}(s)ds$$

Let

$$\dot{y}(t)=\sum_{k=1}^{m}A_k z_k(t)-\sum_{r=1}^{n}(D_r+C_r)l_r(t)+(A-D-C)y(t)$$

where

$$z_k(t)=y(t-d_k(t))-y(t), l_r(t)=y(t-\Gamma_r(t))-y(t)$$

According to the Lemma 4([12]), we have

$$-h_d\sum_{k=1}^{m}\int_{t-h_d}^{t}\dot{y}^T(s)Q_k\dot{y}(s)ds\le-\sum_{k=1}^{m}\left(\int_{t-d_k(t)}^{t}\dot{y}^T(s)ds\right)Q_k\left(\int_{t-d_k(t)}^{t}\dot{y}(s)ds\right)$$
$$=-\sum_{k=1}^{m}\left[y^T(t)-y^T(t-d_k(t))\right]Q_k\left[y(t)-y(t-d_k(t))\right]=-\sum_{k=1}^{m}z_k^T(t)Q_k z_k(t)$$

$$-h_\Gamma\sum_{r=1}^{n}\int_{t-h_\Gamma}^{t}\dot{y}^T(s)R_r\dot{y}(s)ds\le-\sum_{r=1}^{n}\left(\int_{t-\Gamma_r(t)}^{t}\dot{y}^T(s)ds\right)R_r\left(\int_{t-\Gamma_r(t)}^{t}\dot{y}(s)ds\right)$$
$$=-\sum_{r=1}^{n}\left[y^T(t)-y^T(t-\Gamma_r(t))\right]R_r\left[y(t)-y(t-\Gamma_r(t))\right]=-\sum_{r=1}^{n}l_r^T(t)R_r l_r(t)$$

so

$$\dot{V}\le\sum_{k=1}^{m}z_k^T(t)A_k^TPy(t)-\sum_{r=1}^{n}l_r^T(t)G_rPy(t)$$
$$+y^T(t)P\sum_{k=1}^{m}A_k z_k(t)-y^T(t)P\sum_{r=1}^{n}G_r l_k(t)$$
$$+y^T(t)W^TPy(t)+y^T(t)PWy(t)$$
$$+\left[\sum_{k=1}^{m}z_k^T(t)A_k^T-\sum_{r=1}^{n}l_r^T(t)G_r^T+y^T(t)W^T\right](-\Psi)$$
$$\left[\sum_{k=1}^{m}A_k z_k(t)-\sum_{r=1}^{n}G_r l_r(t)+Wy(t)\right]$$
$$-\sum_{k=1}^{m}z_k^T(t)Q_k z_k(t)-\sum_{r=1}^{n}l_r^T(t)R_r l_r(t)$$
$$=\begin{bmatrix}y(t)\\z_1(t)\\\vdots\\z_m(t)\\l_1(t)\\\vdots\\l_n(t)\end{bmatrix}^T\left(\begin{bmatrix}\Theta & PA_1 & \cdots & PA_m & -PG_1 & \cdots & -PG_n\\ * & -Q_1 & \cdots & 0 & 0 & \cdots & 0\\ \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots\\ * & 0 & \cdots & -Q_m & 0 & \cdots & 0\\ * & 0 & \cdots & 0 & -R_1 & \cdots & 0\\ \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots\\ * & 0 & \cdots & 0 & 0 & \cdots & -R_n\end{bmatrix}+\right.$$
$$\left.\begin{bmatrix}W^T(h_d^2Q+h_\Gamma^2R)\\A_1^T(h_d^2Q+h_\Gamma^2R)\\\vdots\\A_m^T(h_d^2Q+h_\Gamma^2R)\\-G_1^T(h_d^2Q+h_\Gamma^2R)\\\vdots\\-G_n^T(h_d^2Q+h_\Gamma^2R)\end{bmatrix}(-\Psi)^{-1}\begin{bmatrix}W^T(h_d^2Q+h_\Gamma^2R)\\A_1^T(h_d^2Q+h_\Gamma^2R)\\\vdots\\A_m^T(h_d^2Q+h_\Gamma^2R)\\-G_1(h_d^2Q+h_\Gamma^2R)\\\vdots\\-G_n(h_d^2Q+h_\Gamma^2R)\end{bmatrix}^T\right)\begin{bmatrix}y(t)\\z_1(t)\\\vdots\\z_m(t)\\l_1(t)\\\vdots\\l_n(t)\end{bmatrix}$$

where $W=A-D-C$.

Using Schur complement Lemma([12]), (6) can be achieved. Obviously, as long as (6) holds, $\dot{V}(t)<0$. Based on the Lyapunov stability theorem, the error system (7) will be stable at zero solution asymptotically so that the system (5) will achieve consensus asymptotically.

C. Decentralized control of UAV formation

In order to track the desired position, we use the consensus protocol (5) as the decentralized control strategy to achieve the multi-UAV formation control. The control algorithm is:

$$u_i(t)=\dot{r}_i^d+\sum_{j\in N_i}a_{ij}\left(\left(r_j-r_j^d\right)(t-\tau_{ij}(t))-\left(r_i-r_i^d\right)(t-T_i(t))\right)$$
$$-\alpha_i\left(\left(r_i-r_i^d\right)(t-T_i(t))-0\right) \quad (8)$$

Though the information exchange among the UAVs, (8) can make the errors between actual positions and desired

positons converge to zero so that each UAV will fly at its desired path and the formation control will achieved.

IV. SIMULATION RESULTS

Consider four UAVs formation control problem in the two-dimensional coordinates. The communication topology is shown in Fig.2.

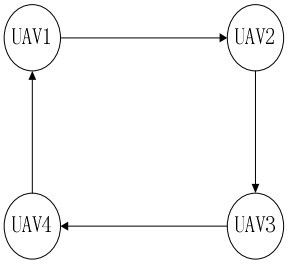


Fig.2. Communication topology of the four UAVs

The adjacency matrix is $A=\begin{bmatrix}0&1&0&0\\0&0&1&0\\0&0&0&1\\1&0&0&0\end{bmatrix}$. Let UAV1 can be access to reference state, that is, $\alpha_1=1$. The communication and self delays are

$$\tau_{12}(t)=d_1(t),\tau_{23}(t)=d_2(t),\tau_{34}(t)=d_3(t),\tau_{41}(t)=d_4(t)$$

,

$$T_1(t)=\Gamma_1(t),T_2(t)=\Gamma_2(t),T_3(t)=\Gamma_3(t),T_4(t)=\Gamma_1(t)$$

Using the Matlab's LMI toolbox to solve LMI (6) , we can get the proper upper bound of communication and self delays. We assume that the communication and self delays are

$$\tau_{12}(t)=0.3|\sin t|+0.1,\tau_{23}(t)=0.2|\sin t|+0.1,$$

$$\tau_{34}(t)=0.1|\sin t|+0.3,\tau_{34}(t)=0.1|\sin t|+0.2$$

$$T_1(t)=0.2|\sin t|,T_2(t)=0.3|\sin t|,$$

$$T_3(t)=0.25|\sin t|,T_4(t)=0.15|\sin t|$$

The formation keeps square. The distances between each UAV and the desired central position are

$$\tilde{r}_{1c}=(-50,50),\tilde{r}_{2c}=(50,50),$$

$$\tilde{r}_{3c}=(50,-50),\tilde{r}_{4c}=(-50,-50)$$

that is, the side length of square is $100\sqrt{2}$. The central desired path is:

$$\dot{x}_c^d=v_c^d\cos(\theta_c^d),\dot{y}_c^d=v_c^d\sin(\theta_c^d),\dot{\theta}_c^d=\omega_c^d$$

All the UAVs set out from $(0,0)$.

A. Broken path

Let $v_c^d=10m/s,\omega_c^d=0,r_c^d(0)=(0,0)$,

$\theta_c^d(t)=\begin{cases}60^\circ & t\le 50s\\ -60^\circ & t>50s\end{cases}$ .The simulation result are as following:

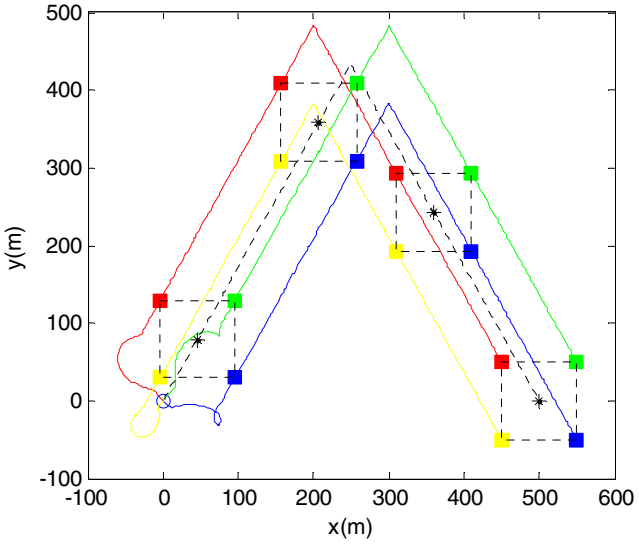


Fig.3. Formation fly at a broken line

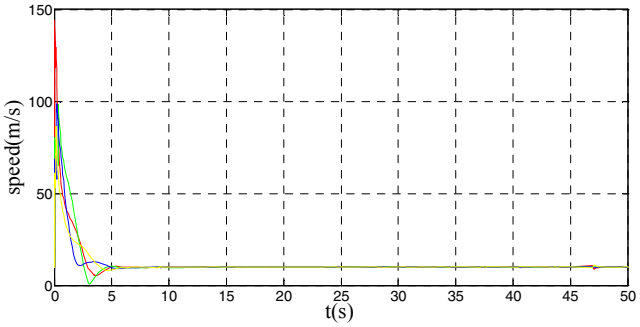


Fig.4. Speed curve

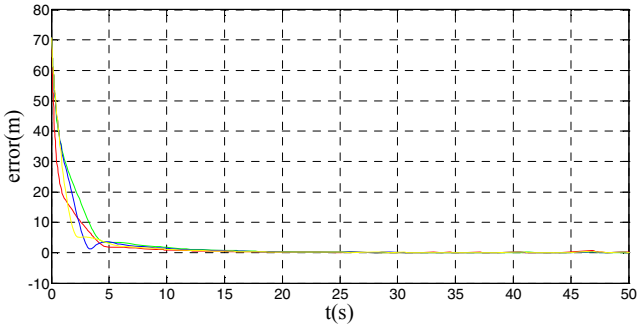


Fig.5. Tracking error

As shown in the Fig 4 and Fig 5, every UAV's tracking errors of factual path and desired path converge to 0 finally and the speeds of UAVs can reach the desired speeds which

demonstrates the multiple UAVs can fly with specified formation.

B. *Spiral path*

Let

$$v_c^d = 0.5t\ m/s, \omega_c^d = \frac{\pi}{20} t\ rad/s, r_c^d(0) = (0,0), \theta_c^d(0) = \frac{\pi}{2}.$$

The simulation results are as following:

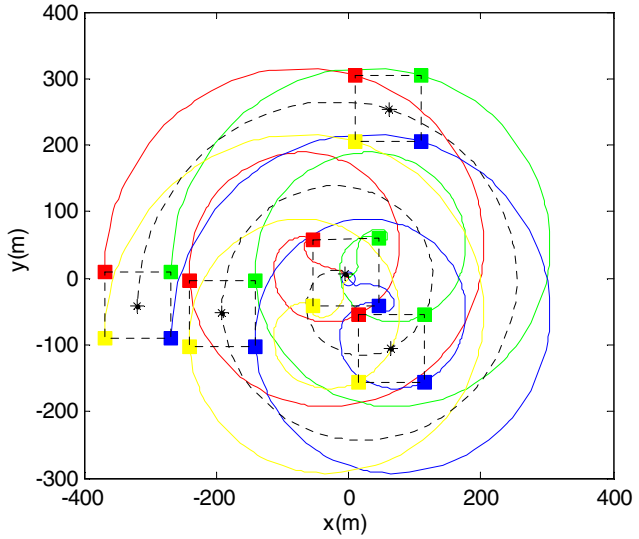


Fig.6. Formation fly at a spiral

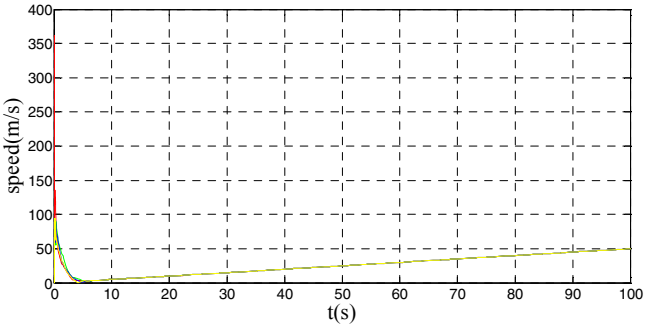


Fig. 7. Speed curve

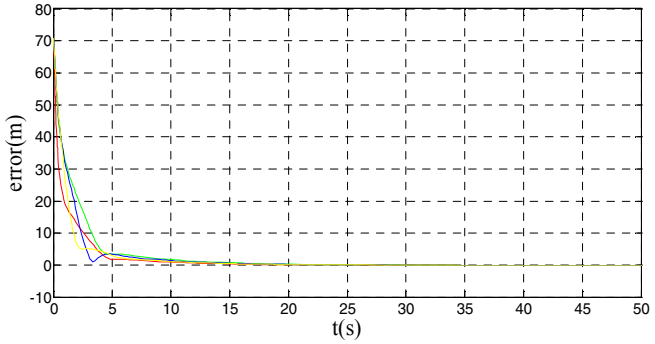


Fig.8. Tracking error

From the Figures above, in the case of existing of communication and self delays, the decentralized control strategy based on leader-following consensus protocol can keep the multi-UAV system formation flying at the desired path.

CONCLUSIONS

This paper put forward a consensus protocol. Combined with decentralized control strategy, the multiple UAVs formation control problem can be solved. The simulation results showed that the system can track the desired path and keep the unchanging of relative positions. Simulation results have been presented to demonstrate the application value.

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